

When Should Actuaries Trust Neural Networks?

A Benchmark of Classical and Neural Mortality Models with a Hybrid LC-ResNet and a Practical Decision Framework

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Abstract

We benchmark 15 mortality forecasting models—8 classical models implemented from scratch, 6 neural network architectures, and an original hybrid—on 8 countries from the Human Mortality Database (1950–2023). Rolling-origin validation with actuarial metrics (life expectancy e_0 , annuity factor $\ddot{a}_{65}$) addresses questions that existing studies leave open: which model to use with short historical data, which is robust to mortality shocks such as COVID-19, and when the added complexity of neural networks is justified. We introduce **LC-ResNet**, a hybrid that couples a Poisson Lee–Carter skeleton with a neural residual correction shrunk by $e^{-\lambda h}$ over the forecast horizon h , preserving long-run stability and interpretability while capturing short-term non-linearities. Key findings: the age-specific random walk with drift outperforms every structural model on log-rate accuracy at all horizons; recurrent κ_t replacements improve on classical Lee–Carter at short horizons but degrade sharply on actuarial metrics at $h = 20$; during the COVID-19 shock, models that do *not* extrapolate a trend are the most robust; and LC-ResNet is the most accurate structural model at $h = 1$ and for young ages. On a 1,000-life French annuity portfolio, the inter-model provision spread reaches 6.3 million EUR, and the Solvency II longevity SCR varies by 7% across models—model choice is a capital decision.

Keywords: Mortality Forecasting · Lee–Carter · Deep Learning · Longevity Risk · Model Risk · Solvency II · Benchmark

JEL: C45 · C52 · C53 · G22 · J11

1 Introduction

Longevity risk—the risk that policyholders live longer than expected—is a primary concern for life insurers and pension funds. Under Solvency II, the capital requirement for longevity risk typically represents 5–15% of the best estimate of annuity provisions, so the choice of mortality model is a capital allocation decision rather than a purely academic one.

The Lee–Carter family (Lee and Carter, 1992) has dominated actuarial practice for three decades. Neural network approaches have shown promising results in recent literature (Richman and Wüthrich, 2021; Nigri et al., 2019; Perla et al., 2021), yet no study offers a practical guide for choosing between classical and neural models given the constraints practitioners actually face: data availability, forecast horizon, age segment, and regulatory interpretability.

This paper fills that gap with four contributions:

1. **A comprehensive benchmark** of 15 models (8 classical, 6 neural, 1 hybrid) on 8 countries, evaluated with statistical *and* actuarial metrics under rolling-origin validation (Sections 4–6).
2. **LC-ResNet**, an original hybrid coupling a Poisson Lee–Carter skeleton with a neural residual network under horizon-dependent shrinkage (Section 4.3).
3. **A decision framework** mapping history length, horizon, age segment, and interpretability needs to a recommended model (Section 7).

4. **An actuarial case study** pricing a French annuity portfolio and quantifying model risk in EUR under a Solvency II longevity shock (Section 8).

Figure 1 summarizes the benchmark pipeline.

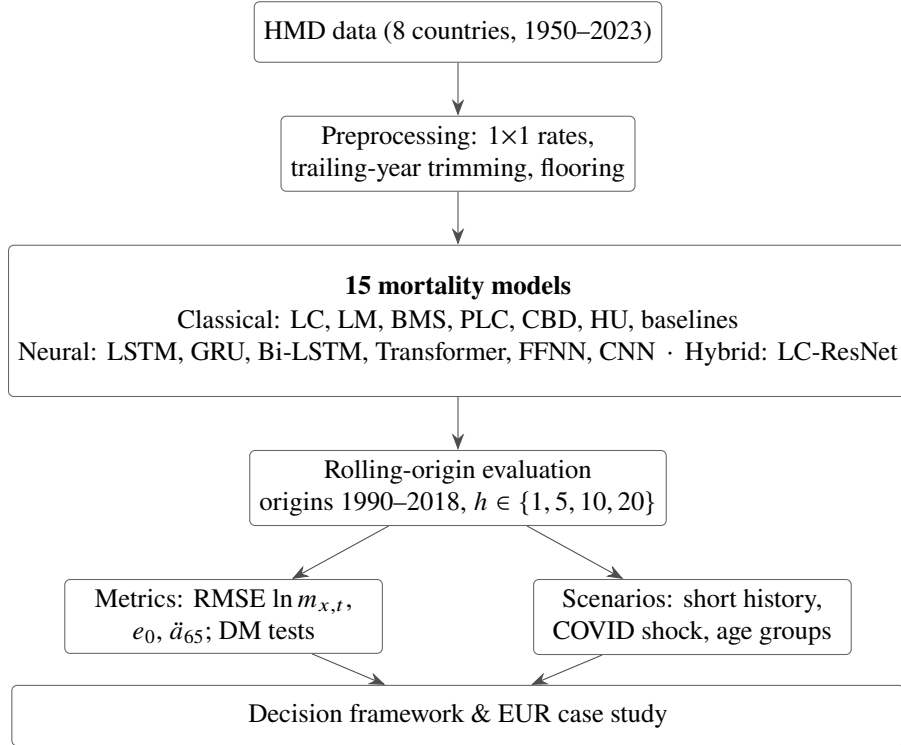


Figure 1: Benchmark pipeline: from raw HMD data to the decision framework and the actuarial case study.

2 Related Work

Lee and Carter (1992) introduced the seminal bilinear model for U.S. mortality. Extensions include the e_0 -adjusted variant of Lee and Miller (2001), the fitting-period selection of Booth et al. (2002), the Poisson likelihood formulation of Brouhns et al. (2002), the two-factor CBD model for older ages (Cairns et al., 2006), and the functional data approach of Hyndman and Ullah (2007). Booth (2006) review this literature and document the surprising strength of naïve extrapolation.

On the neural side, Nigri et al. (2019) replace the random-walk forecast of κ_t with an LSTM; Richman and Wüthrich (2021) propose a feed-forward network with entity embeddings for multi-population mortality; Perla et al. (2021) treat mortality forecasting as time-series regression with convolutional and recurrent architectures.

Our work differs in three ways: (i) all approaches are benchmarked under identical conditions—same data, same rolling origins, same hyperparameter budget; (ii) evaluation includes actuarial metrics and an EUR-denominated case study rather than log-rate accuracy alone; (iii) we propose a hybrid model and a practical decision framework.

3 Data

We use 1×1 central death rates $m_{x,t}$ and exposures $E_{x,t}$ from the Human Mortality Database (HMD)¹ for eight countries: France, England & Wales, USA, Japan, Italy, Spain, Sweden, and the Netherlands. Ages

¹<https://www.mortality.org>. HMD data are free but not redistributable; the repository documents how to download them.

span 0–100, years 1950–2023 (total population). The window includes the COVID-19 shock (2020–2023), used as a natural out-of-sample stress test.

The central death rate is defined as

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}}, \quad (1)$$

where $D_{x,t}$ denotes deaths at age x in year t and $E_{x,t}$ the matching central exposure. Zero or missing rates are floored at 10^{-6} before taking logarithms. Countries whose series end before 2023 (e.g. England & Wales, last available year 2022) have trailing missing years trimmed rather than imputed.

4 Models

All classical models are implemented from scratch in Python (NumPy/SciPy); neural models use PyTorch. Every model implements a common interface (`fit`, `forecast`, `simulate`), so the evaluation harness treats all 15 identically.

4.1 Classical models

Lee–Carter (LC). The bilinear structure

$$\ln m_{x,t} = a_x + b_x \kappa_t + \varepsilon_{x,t}, \quad (2)$$

estimated by SVD under the constraints $\sum_x b_x = 1$ and $\sum_t \kappa_t = 0$, with κ_t re-estimated on observed death counts. The index κ_t is forecast by a random walk with drift.

Lee–Miller (LM). Adjusts κ_t each year so that the model-implied life expectancy matches the observed e_0 (Lee and Miller, 2001).

Booth–Maindonald–Smith (BMS). Selects the fitting period by testing the linearity of κ_t , then applies LC on the selected window (Booth et al., 2002).

Poisson Lee–Carter (PLC). Assumes $D_{x,t} \sim \text{Poisson}(E_{x,t} e^{a_x + b_x \kappa_t})$ and maximizes the likelihood by alternating Newton–Raphson updates (Brouhns et al., 2002).

Cairns–Blake–Dowd (CBD). For ages 60–100,

$$\text{logit } q_{x,t} = \kappa_t^{(1)} + \kappa_t^{(2)} (x - \bar{x}), \quad (3)$$

with a bivariate random walk with drift on $(\kappa_t^{(1)}, \kappa_t^{(2)})$ (Cairns et al., 2006).

Hyndman–Ullah (HU). Penalized-spline smoothing of the age profiles, functional principal component decomposition ($K = 6$), and ARIMA forecasting of the scores (Hyndman and Ullah, 2007).

Baselines. An age-specific random walk with drift on $\ln m_{x,t}$, and frozen rates (last observed year held constant). Honest benchmarks require both.

4.2 Neural models

For benchmark fairness, every neural model uses the same budget: 200 epochs, early stopping with patience 20, Adam with learning rate 10^{-3} and weight decay 10^{-4} , fixed seed. Architectures are detailed in Appendix A.

Neural κ_t models. The LC skeleton (a_x, b_x) is retained; only the κ_t forecast is replaced by an LSTM, GRU, Bi-LSTM, or a two-layer Transformer encoder operating on a 20-year lookback window.

FFNN with embeddings. Following Richman and Wüthrich (2021): an age embedding concatenated with a continuous normalized year feature, mapped by a three-layer perceptron directly to $\ln m_{x,t}$. The continuous year encoding—rather than a year embedding—is what allows extrapolation to unseen future years.

CNN on the mortality surface. The $(\text{age} \times \text{year})$ matrix is treated as an image; three convolutional layers predict the next annual profile from a sliding window, applied recursively for multi-step forecasts.

4.3 LC-ResNet: a hybrid with horizon shrinkage

Our proposed hybrid preserves the interpretable LC skeleton and lets a small network correct only what the skeleton misses:

1. Fit a Poisson Lee–Carter, obtaining $\hat{a}_x, \hat{b}_x, \hat{\kappa}_t$.
2. Compute the structured residuals $r_{x,t} = \ln m_{x,t} - (\hat{a}_x + \hat{b}_x \hat{\kappa}_t)$.
3. Train a residual network f_θ (MLP 64–32) on normalized features $(\tilde{x}, \tilde{t}, \tilde{b}_x, \tilde{\kappa}_t)$ to predict $r_{x,t}$.
4. Forecast with horizon-dependent shrinkage:

$$\widehat{\ln m_{x,T+h}} = \underbrace{\hat{a}_x + \hat{b}_x \hat{\kappa}_{T+h}}_{\text{LC skeleton}} + e^{-\lambda h} \underbrace{f_\theta(\tilde{x}, \tilde{t}+h, \tilde{b}_x, \tilde{\kappa}_{T+h})}_{\text{neural correction}}, \quad \lambda = 0.1. \quad (4)$$

At $h = 1$ the correction is almost fully active ($e^{-0.1} \approx 0.90$); at $h = 20$ it has decayed to $e^{-2} \approx 0.14$ and the model reverts to the stable LC extrapolation. This directly targets the main failure mode of neural mortality models—long-horizon extrapolation instability—while (a_x, b_x, κ_t) remain available for regulatory reporting.

5 Evaluation Protocol

Rolling-origin validation. For each origin $T_0 \in \{1990, 1992, \dots, 2018\}$, every model is re-trained from scratch on years $[1950, T_0]$ and evaluated on the following $h \in \{1, 5, 10, 20\}$ years, whenever the truth is available.

Metrics. RMSE on $\ln m_{x,t}$; RMSE on period life expectancy at birth e_0 (in years); RMSE on the whole-life annuity-due factor $\ddot{a}_{65}$ at interest rate $i = 2\%$. Actuarial metrics require the full age grid, so CBD (ages 60+) is evaluated on log-rates only.

Significance. Pairwise Diebold–Mariano tests (Diebold and Mariano, 1995) with the Harvey–Leybourne–Newbold small-sample and multi-step correction (Harvey et al., 1997), referenced against a Student- t distribution.

Scenarios. Three targeted experiments feed the decision framework: (i) *short history*—training truncated to the last 20, 30, or 50 years before the origin; (ii) *mortality shock*—training up to 2019, evaluation on 2020–2022 (COVID-19); (iii) *age groups*—separate evaluation on ages 0–19, 20–64, and 65–100.

6 Results

The benchmark yields 29,330 result rows (model $\times$ country $\times$ origin $\times$ horizon $\times$ metric). Tables report cross-country averages; per-country files are in the repository. Pre-COVID rolling origins ($T_0 < 2019$) and the COVID shock are reported separately so that neither contaminates the other.

6.1 Accuracy on log death rates

Key findings.

1. The **age-specific random walk with drift dominates every structural model** on log-rate accuracy at every horizon (0.100 at $h=1$ to 0.203 at $h=20$)—a sobering baseline result consistent with Booth (2006).
2. Among structural models, **LC-ResNet is the most accurate at $h=1$** (0.120) and stays competitive throughout, matching the LSTM/GRU at $h=5$ and beating every neural model at $h=20$ (0.247), where the shrinkage has handed control back to the LC skeleton.

Table 1: RMSE on $\ln m_{x,t}$ by forecast horizon, averaged over 8 countries and all pre-COVID rolling origins. Lower is better; best full-grid model per column in bold. CBD forecasts ages 60–100 only, so its values are not comparable to full-grid models.

Model	$h=1$	$h=5$	$h=10$	$h=20$
<i>Classical</i>				
Lee–Carter	0.156	0.177	0.199	0.248
Lee–Miller	0.145	0.166	0.187	0.232
BMS	0.156	0.177	0.199	0.248
Poisson LC	0.190	0.214	0.231	0.272
CBD (60+ only)	0.076	0.087	0.095	0.120
Hyndman–Ullah	0.316	0.322	0.329	0.355
<i>Baselines</i>				
Random walk	0.100	0.124	0.146	0.203
Frozen rates	0.100	0.133	0.186	0.314
<i>Neural</i>				
LSTM κ_t	0.131	0.152	0.180	0.263
GRU κ_t	0.130	0.152	0.182	0.274
Bi-LSTM κ_t	0.132	0.157	0.195	0.288
Transformer κ_t	0.141	0.177	0.225	0.327
FFNN embeddings	0.231	0.257	0.290	0.367
CNN surface	0.147	0.191	0.240	0.448
<i>Hybrid</i>				
LC-ResNet	0.120	0.152	0.184	0.247

3. **Recurrent κ_t models** (LSTM, GRU) beat classical LC at $h \leq 10$ but lose that edge at $h=20$; the **Transformer and CNN degrade most** at long horizons (0.327 and 0.448), the classic extrapolation failure of flexible models.
4. **Frozen rates collapse with horizon** (0.100 $\rightarrow$ 0.314), as expected for a no-trend baseline: mortality does improve over 20 years.

6.2 Actuarial accuracy

Table 2: Actuarial metrics at $h=5$ and $h=20$: RMSE on life expectancy at birth e_0 (years) and on the annuity factor $\ddot{a}_{65}$. Averaged over 8 countries, pre-COVID origins. Best per column in bold.

Model	RMSE e_0 (years)		RMSE $\ddot{a}_{65}$	
	$h=5$	$h=20$	$h=5$	$h=20$
Lee–Carter	0.428	0.832	0.178	0.397
Lee–Miller	0.318	0.742	0.195	0.398
Poisson LC	0.376	0.826	0.181	0.403
Hyndman–Ullah	0.419	0.694	0.188	0.437
Random walk	0.315	0.764	0.159	0.395
Frozen rates	0.607	2.417	0.272	1.096
LSTM κ_t	0.739	2.059	0.399	1.017
GRU κ_t	0.749	2.181	0.397	1.065
Transformer κ_t	1.202	2.824	0.596	1.337
FFNN embeddings	0.557	0.787	0.316	0.381
LC-ResNet	0.353	0.846	0.174	0.403

The central observation is an **inversion between statistical and actuarial accuracy for recurrent models**: the LSTM beats Lee–Carter on log-rates at $h \leq 10$ (Table 1) yet its e_0 error at $h=20$ is 2.06

years against 0.74 for Lee–Miller. Life expectancy integrates the entire mortality surface, so small but systematic biases in the κ_t path compound through the life table. LC-ResNet avoids this failure: its shrinkage returns the long-horizon forecast to the LC skeleton, keeping e_0 and $\ddot{a}_{65}$ errors at classical levels.

6.3 COVID-19 shock

Table 3: COVID-19 stress test: models trained through 2019, evaluated on 2020–2022. RMSE on $\ln m_{x,t}$; best full-grid model per column in bold.

Model	2020	2021	2022
Frozen rates	0.149	0.153	0.152
LSTM κ_t	0.150	0.156	0.158
Bi-LSTM κ_t	0.150	0.154	0.155
Transformer κ_t	0.150	0.154	0.153
LC-ResNet	0.152	0.167	0.183
Random walk	0.153	0.164	0.173
CBD (60+ only)	0.154	0.147	0.146
Lee–Miller	0.170	0.183	0.195
Lee–Carter	0.180	0.193	0.205
Poisson LC	0.236	0.252	0.265
Hyndman–Ullah	0.336	0.337	0.337

Under the shock, the ranking flips: **models that extrapolate no trend (frozen rates) or a dampened trend are the most robust**, because COVID temporarily erased the improvement trend that LC-type models kept projecting. Neural κ_t models sit close behind frozen rates. This is a general lesson: trend-following accuracy in calm periods and shock robustness are different competencies, and no model offers both.

6.4 Short history and age segments

With training truncated to 20 years, LC-ResNet is the best full-grid structural model (RMSE 0.159 versus 0.165 for LC and 0.173 for the random walk); frozen rates again lead among baselines (0.134). For **young ages (0–19)**, LC-ResNet ranks first overall (0.228), suggesting the residual network captures infant- and child-mortality non-linearities that the bilinear form misses. For **elderly ages (65–100)**—the annuity segment—simple approaches win: frozen rates (0.054), Hyndman–Ullah and the random walk (0.068). Full tables appear in Appendix B.

7 Decision Framework

Table 4 distills the benchmark into model selection guidance. The overarching message: **no single model dominates**; the choice should follow the use case, and the burden of proof is on complexity.

Neural κ_t replacements are justified only when the history is long (≥ 50 years), the horizon short (≤ 5 years), and log-rate accuracy—not e_0 or annuity accuracy—is the objective. Outside that envelope their added complexity buys little or is actively harmful (Table 2).

8 Actuarial Case Study

Setup. A portfolio of 1,000 French pensioners aged 65, each receiving 12,000 EUR annually; discount rate $i = 2\%$; mortality from each model’s 10-year-ahead forecast on French data.

Table 4: Which model to use when. Recommendations follow directly from the scenario results.

Context	Recommendation	Evidence
Short history (≤ 25 yr)	LC-ResNet	Best structural model at hist-20 (Table 9)
Shock regime / pandemic	Frozen rates, neural κ_t	Table 3: no-trend models most robust
Long horizon (> 10 yr), provisioning	Lee–Miller	Best structural $\ln m$ and e_0 at $h=20$
Short horizon (≤ 5 yr)	LC-ResNet	Best structural model at $h=1$, ties LSTM at $h=5$
Elderly segment, regulatory	CBD	Purpose-built for 60+, regulator-standard
Quick unbiased reference	Random walk	Best log-rate accuracy at every horizon
Interpretability required	LC-ResNet or LC family	(a_x, b_x, κ_t) fully reportable

Table 5: Annuity pricing by model. The provision is $1,000 \times 12,000 \times \ddot{a}_{65}$.

Model	$\ddot{a}_{65}$	Provision / person (EUR)	Total (MEUR)
Lee–Carter	18.338	220,050	220.05
Poisson LC	18.323	219,873	219.87
LC-ResNet	18.268	219,217	219.22
Random walk	18.236	218,826	218.83
CBD	17.816	213,793	213.79
Inter-model spread			6.26

The provision spread across models reaches **6.26 MEUR** (2.8% of the largest provision), driven mainly by CBD, which projects higher old-age mortality; among full-grid models the spread is 1.22 MEUR. Either way, the differences are material at portfolio scale.

Table 6: Solvency II longevity shock (-20% on mortality rates).

Model	Base (MEUR)	Shocked (MEUR)	SCR (MEUR)	SCR (%)
Lee–Carter	220.05	231.57	11.52	5.23
Poisson LC	219.87	231.39	11.52	5.24
LC-ResNet	219.22	230.80	11.58	5.28
Random walk	218.83	230.54	11.71	5.35
CBD	213.79	226.12	12.33	5.77

The longevity SCR ranges from 11.52 to 12.33 MEUR—a 7% relative spread—while the SCR-to-provision ratio is stable (5.2–5.8%). Model choice therefore shifts both the best estimate and the regulatory capital, but the shock multiplier itself is robust to specification.

9 Discussion

The random walk result. The strongest single finding is that an age-specific random walk with drift beats every structural model on log-rate accuracy at every horizon. Structure helps not by improving raw accuracy but by improving *actuarial* accuracy ($e_0, \ddot{a}_{65}$), where cross-age consistency matters—see the contrast between Tables 1 and 2.

When neural networks help, and how they fail. Recurrent κ_t models genuinely improve short-horizon log-rate accuracy, and they adapt quickly during the COVID shock. Their failure mode is systematic long-horizon bias that compounds through the life table: a 2-year e_0 error at $h=20$ is disqualifying for provisioning. Model complexity must be matched to the decision the forecast serves.

Bounded flexibility works. LC-ResNet’s design principle—let a neural network correct the skeleton where evidence is dense (short horizons, young ages) and fade it out where extrapolation risk dominates—

delivers the intended profile: best structural model at $h=1$ and for young ages, classical-level actuarial accuracy at $h=20$, and full interpretability of the skeleton.

Limitations. All models are single-population; cohort effects are not explicitly modeled; neural results use a single seed rather than deep ensembles; prediction-interval calibration is not evaluated here; and post-2023 data will be needed to judge post-COVID trend re-establishment.

10 Conclusion

When should actuaries trust neural networks for mortality forecasting? On this evidence: at short horizons, with long training histories, for log-rate objectives—and even then, preferably inside a constrained design such as LC-ResNet, where the neural component is bounded and fades with horizon. For long-horizon provisioning, the classical family (Lee–Miller in particular) remains the safer instrument, and the random walk with drift should always be run as the honesty check. The 6.3 MEUR provision spread on a modest portfolio shows that these choices are, ultimately, capital decisions.

Code Availability

The complete implementation—all 15 models from scratch, the evaluation harness, the pricing case study, tests, and an interactive dashboard—is openly available at:

<https://github.com/aminemanai2003/neural-mortality-benchmark>

HMD data are not redistributable; the repository documents the download procedure. All results are reproducible with `python scripts/run_benchmark.py` (fixed seeds).

References

- Booth, H. (2006). Demographic forecasting: 1980 to 2005 in review. *International Journal of Forecasting*, 22(3):547–581.
- Booth, H., Maindonald, J., and Smith, L. (2002). Applying Lee-Carter under conditions of variable mortality decline. *Population Studies*, 56(3):325–336.
- Brouhns, N., Denuit, M., and Vermunt, J. K. (2002). A Poisson log-bilinear regression approach to the construction of projected lifetables. *Insurance: Mathematics and Economics*, 31(3):373–393.
- Cairns, A. J. G., Blake, D., and Dowd, K. (2006). A two-factor model for stochastic mortality with parameter uncertainty: Theory and calibration. *Journal of Risk and Insurance*, 73(4):687–718.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.
- Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.
- Hyndman, R. J. and Ullah, M. S. (2007). Robust forecasting of mortality and fertility rates: A functional data approach. *Computational Statistics & Data Analysis*, 51(10):4942–4956.
- Lee, R. D. and Carter, L. R. (1992). Modeling and forecasting U.S. mortality. *Journal of the American Statistical Association*, 87(419):659–671.
- Lee, R. D. and Miller, T. (2001). Evaluating the performance of the Lee-Carter method for forecasting mortality. *Demography*, 38(4):537–549.

- Nigri, A., Levantesi, S., Marino, M., Scognamiglio, S., and Perla, F. (2019). A deep learning integrated Lee-Carter model. *Risks*, 7(1):33.
- Perla, F., Richman, R., Scognamiglio, S., and Wüthrich, M. V. (2021). Time-series forecasting of mortality rates using deep learning. *Scandinavian Actuarial Journal*, 2021(7):572–598.
- Richman, R. and Wüthrich, M. V. (2021). A neural network extension of the Lee-Carter model to multiple populations. *Annals of Actuarial Science*, 15(2):346–366.

A Model Configurations

Table 7: Hyperparameters. All neural models: 200 epochs max, early stopping (patience 20), Adam, learning rate 10^{-3} , weight decay 10^{-4} , seed 42.

Model	Architecture
LSTM / GRU / Bi-LSTM κ_t	2 layers, hidden 64, lookback 20
Transformer κ_t	2 encoder layers, $d_{\text{model}}=64$, 4 heads, lookback 20
FFNN embeddings	age emb. dim 8 + continuous year; MLP 128–64–32, dropout 0.1
CNN surface	channels 16–32–64, kernel 3×3, adaptive pooling
LC-ResNet	residual MLP 64–32, dropout 0.1, $\lambda = 0.1$
CBD	age range 60–100
Hyndman–Ullah	6 functional components, penalized splines

Table 8: Mean training time per fit (seconds, single CPU).

Model	Time (s)	Model	Time (s)
Frozen rates	0.00	Bi-LSTM κ_t	0.47
Random walk	0.00	Transformer κ_t	0.55
CBD	0.00	GRU κ_t	0.73
Hyndman–Ullah	0.00	LC-ResNet	1.45
Lee–Carter	0.01	FFNN embeddings	3.27
BMS	0.03	CNN surface	8.18
Poisson LC	0.05		
Lee–Miller	0.09		

B Additional Results

Table 9: Short-history scenario: RMSE on $\ln m_{x,t}$ with training truncated to the last 20 or 50 years (averaged over countries and horizons $h \in \{5, 10\}$). Best full-grid model in bold.

Model	20 yr	50 yr
CBD (60+ only)	0.128	0.122
Frozen rates	0.134	0.134
LC-ResNet	0.159	0.151
Poisson LC	0.163	0.189
Lee–Miller	0.164	0.163
Lee–Carter / BMS	0.165	0.163
LSTM κ_t	0.165	0.157
Bi-LSTM κ_t	0.165	0.147
GRU κ_t	0.165	0.156
Transformer κ_t	0.165	0.164
Random walk	0.173	0.149
CNN surface	0.178	0.220
FFNN embeddings	0.290	0.255
Hyndman–Ullah	0.340	0.333

Table 10: Age-group evaluation: RMSE on $\ln m_{x,t}$ for young (0–19) and elderly (65–100) ages, averaged over countries, origins, and horizons. Best per column in bold.

Model	Young 0–19	Elderly 65–100
LC-ResNet	0.228	0.070
Bi-LSTM κ_t	0.228	0.071
LSTM κ_t	0.229	0.074
GRU κ_t	0.230	0.074
Lee–Carter / BMS	0.233	0.079
Lee–Miller	0.236	0.080
Frozen rates	0.241	0.054
Transformer κ_t	0.246	0.078
FFNN embeddings	0.252	0.117
Random walk	0.263	0.068
Hyndman–Ullah	0.346	0.068
Poisson LC	0.314	0.073
CNN surface	0.332	0.125
CBD (60+ only)	–	0.102